

ATIVIDADE DE APOIO – ENSINO MÉDIO

Componente Curricular: Física

Professor (a): Rosana

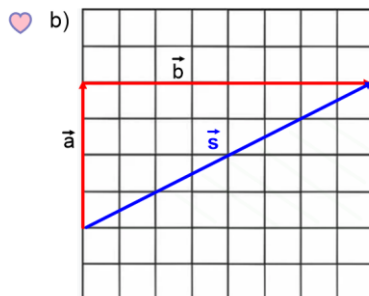
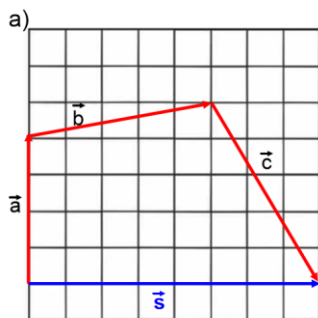
Assunto (s): Adição de Vetores – Livro – Exercícios - Resolução

Data: Abril

Série: 1

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1- Considerar que os lados de cada quadradinho mede 1 u.m.

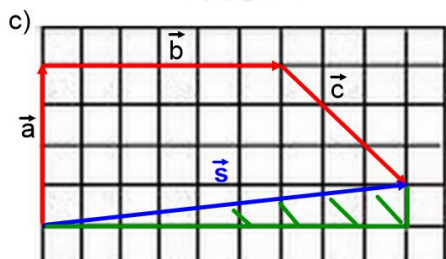


♥ $|\vec{s}| = 8 \text{ u.m.}$

♥ $|\vec{s}|^2 = 4^2 + 8^2 \Rightarrow |\vec{s}|^2 = 16 + 64$

$|\vec{s}| = \sqrt{80}$

$|\vec{s}| = \sqrt{2^4 \cdot 5} \Rightarrow |\vec{s}| = 4\sqrt{5} \text{ u.m.}$

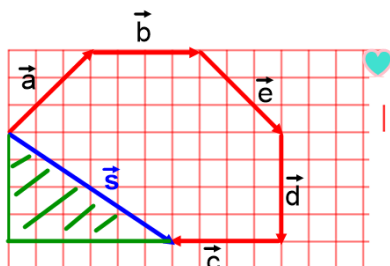


♥ $|\vec{s}|^2 = 9^2 + 1^2 \Rightarrow |\vec{s}|^2 = 81 + 1$

$|\vec{s}| = \sqrt{90}$

$|\vec{s}| = \sqrt{2 \cdot 3^2 \cdot 5} \Rightarrow |\vec{s}| = 3\sqrt{10} \text{ u.m.}$

2-



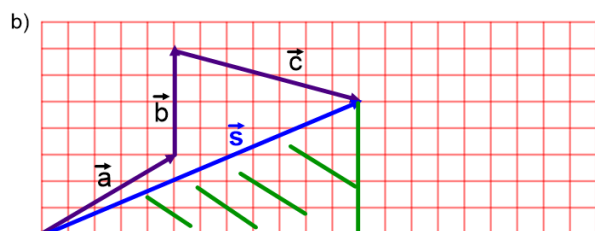
♥ $|\vec{s}|^2 = 4^2 + 6^2 \Rightarrow |\vec{s}|^2 = 16 + 36$

$|\vec{s}| = \sqrt{52}$

$|\vec{s}| = \sqrt{2^2 \cdot 13} \Rightarrow |\vec{s}| = 2\sqrt{13} \text{ u.m.}$

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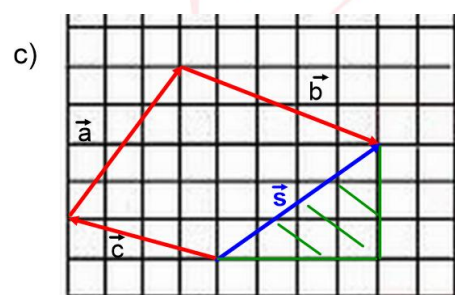
Considerar que os lados de cada quadradinho mede 1 cm



$$|\vec{s}|^2 = 12^2 + 5^2 \Rightarrow |\vec{s}|^2 = 144 + 25$$

$$|\vec{s}| = \sqrt{169}$$

$$|\vec{s}| = 13 \text{ cm}$$

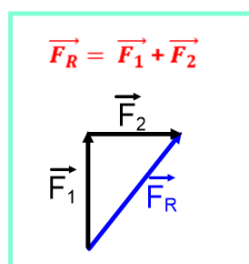
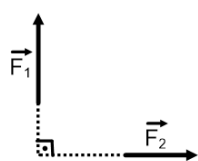


$$|\vec{s}|^2 = 3^2 + 4^2 \Rightarrow |\vec{s}|^2 = 9 + 16$$

$$|\vec{s}| = \sqrt{25}$$

$$|\vec{s}| = 5 \text{ cm}$$

2- **Força Resultante ($\vec{F}_R$) soma dos vetores forças dados**



$$F_R = 75 \text{ N}$$

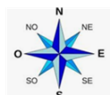
$$F_1 = 60 \text{ N}$$

$$F_2 = ?$$

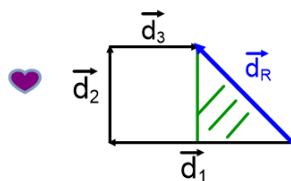
$$|\vec{F}_R|^2 = |\vec{F}_1|^2 + |\vec{F}_2|^2 \Rightarrow 75^2 = 60^2 + |\vec{F}_2|^2 \Rightarrow 5625 = 3600 + |\vec{F}_2|^2$$

$$|\vec{F}_2|^2 = 2025 \Rightarrow |\vec{F}_2| = 45 \text{ N}$$

3-



$$\vec{d}_R = \vec{d}_1 + \vec{d}_2 + \vec{d}_3$$



$$*d_1=12 \text{ m}; d_2=8 \text{ m}; d_3=6 \text{ m}$$

$$|\vec{d}_R|^2 = 6^2 + 8^2 \Rightarrow |\vec{d}_R|^2 = 36 + 64 \Rightarrow |\vec{d}_R|^2 = 100 \Rightarrow |\vec{d}_R| = \sqrt{100} \Rightarrow |\vec{d}_R| = 10 \text{ m}$$